A Declared-Truncation Reconstruction of a 41-Coupling Gravity–Standard-Model FRG System: Verified Blocks, Joint Fixed Point, and Stability Spectrum

C. M. Baird Zora

October 13, 2026

Abstract

We reconstruct, in a fully declared and auditable manner, the running-coupling system of a four-dimensional functional-renormalization-group (FRG) truncation that couples quantum gravity to Standard-Model matter and two real gauge-singlet scalars at mass dimension $d \leq 6$. The original archival claim of a 41-running-coupling ultraviolet completion is not accompanied by an operator list, a beta-functional system, a fixed-point vector, or a stability matrix; rather than fabricating the missing input, we declare a 41-coordinate basis, record a regulator, gauge, and projection convention for every block, and derive only those flows that trace to a written source or a documented projection. Thirty-three of the forty-one beta functions are thereby supplied and verified: the Einstein–Hilbert–matter block of Donà, Eichhorn and Percacci; the nonminimal scalar–curvature system of Narain and Percacci; the scalar local-potential block at quartic and sixth order; and the curvature-squared block of Codello, Percacci and Rahmede. The joint fixed point of the derived flow is the DEP Standard-Model fixed point $(\lambda_N, g_N) = (-4.7975, 1.7621)$ with Gaussian matter. The 33-dimensional quantum-plus-canonical stability spectrum contains 8 relevant, 12 marginal, and 13 irrelevant directions; the two gravity exponents reproduce the published values $(3.961, 1.644)$. We also record a negative result: in the one-loop curvature-squared block the scaffold coordinate of the Weyl-squared coupling runs monotonically, so that block admits no fixed point in that parametrization. Six derivative and gauge-portal operators remain undriven and are explicitly declared pending.

1 Introduction

The asymptotic-safety scenario for quantum gravity [3, 5, 9] is one of the few quantum-gravity proposals that can be addressed quantitatively. Within the functional renormalization group, the central object is the effective average action Γ_k, whose scale dependence is governed by the exact Wetterich equation [1, 2],

$$\partial_t \Gamma_k[\Phi] = \frac{1}{2}\,\mathrm{STr}\left[\left(\Gamma_k^{(2)}[\Phi] + R_k\right)^{-1} \partial_t R_k\right], \tag{1}$$

where $\Gamma_k^{(2)}$ is the second functional derivative (Hessian), R_k a momentum cutoff (regulator), and $t = \ln(k/k_0)$ the logarithmic scale parameter. In practice the equation is solved in truncated theory spaces, and every numerical claim carries a hidden load of choices: the operator basis, the regulator, the gauge fixing and ghost sector, and the projection background.

This paper is the scientific write-up of a reconstruction exercise that began from a corpus claiming a 41-running-coupling, dimension-$d \leq 6$ truncation coupling gravity to the Standard Model

plus two scalar singlets. The claim was not accompanied by the mathematical objects needed to reproduce it: no complete operator list, no beta functions, no fixed-point vector, and no 41×41 stability matrix. The reconstruction therefore proceeds under a strict *no-fabrication* protocol: every beta function must trace to a written source equation (with the paper, equation, gauge, and regulator recorded) or to an explicitly declared projection of a documented trace functional. Coordinates without a traced flow are marked `NaN` (not set to zero), and the framework refuses to present the result as the historical 41-coupling theory. Instead it delivers a smaller, fully declared truncation whose content is *verified*, together with an honest inventory of what remains undriven.

The main results of this paper are:

1. a declared 41-coordinate basis with recorded conventions (Sec. 2, App. A);
2. five verified building blocks, each checked against its published source (Sec. 3);
3. the joint fixed point of the derived 33-coordinate flow and its stability spectrum, with 8 relevant, 12 marginal, and 13 irrelevant directions (Sec. 4);
4. a recorded negative result in the curvature-squared sector (Sec. 3.6, Sec. 6);
5. an explicit list of the six pending operators that remain undriven (Sec. 5).

Throughout, the numbers quoted are produced by the accompanying Python implementation, which is exercised by a unit-test suite; every table in this paper corresponds to a test or a fixed-point verification in that suite.

2 Framework, conventions, and controls

2.1 Fields, action, and the Wetterich equation

We work in Euclidean signature in $d = 4$ dimensions with a background-field formulation. The field content is the metric $g_{\mu\nu}$, the Standard-Model fields, and two real gauge-singlet scalars Φ_c and E. The flow equation is the Wetterich equation (1); the Hessian is constructed covariantly, background-covariant gauge fixing with Faddeev–Popov ghosts is used, and a specific regulator and projection background must be recorded with every numerical result. The implementation adopts the optimized (Litim) shape [4],

$$R_k(q^2) = \big(k^2 - q^2\big)\,\theta\big(k^2 - q^2\big)\,, \tag{2}$$

except in the gravity–matter block of Sec. 3.1, which uses the type-II cutoff of its source paper; the two regulator families are never mixed in one solve.

2.2 The declared 41-coordinate basis

The coordinates follow `model.NAMES` in order:

1. g_N: dimensionless Newton coupling (Einstein–Hilbert, R).
2. λ_N: dimensionless cosmological term.
3. a_{R^2}, a_{C^2}: R^2 and Weyl-squared coordinates.
4. –7. g_1, g_2, g_3: Standard-Model gauge couplings.
5. –10. y_t, y_b, y_τ: third-generation Yukawas.
6. –16. λ_H, m_H^2; λ_Φ, m_Φ^2; λ_E, m_E^2: quartics and masses of the Higgs and the two singlets.
7. –19. $\kappa_{\Phi E}$, $\kappa_{H\Phi}$, κ_{HE}: quartic portals $\Phi_c^2 E^2$, $|H|^2\Phi_c^2$, $|H|^2E^2$.
8. –22. ξ_H, ξ_Φ, ξ_E: nonminimal couplings $R|H|^2$, $R\Phi_c^2$, RE^2.
9. –25. Z_h, Z_ϕ, Z_E: wave-function renormalizations.
10. –35. ten scalar dimension-six potentials: $|H|^6$, Φ^6, E^6, $|H|^4\Phi^2$, $|H|^4E^2$, $|H|^2\Phi^4$, $|H|^2E^4$, Φ^4E^2, Φ^2E^4, $|H|^2\Phi^2E^2$ (implementation names $c_{H6} \ldots c_{H2\Phi2E2}$).

11. –38. derivative dimension-six operators: $|H|^2\Box|H|^2$, $\Phi^2\Box\Phi^2$, $E^2\Box E^2$ ($c_{H\Box}, c_{\Phi\Box}, c_{E\Box}$).
12. –41. gauge-strength portals $|H|^2F^2$, Φ^2F^2, E^2F^2 ($c_{HF^2}, c_{\Phi F^2}, c_{EF^2}$).
The full table with physical definitions is collected in App. A.

2.3 Canonical scaling and the no-fabrication control

Every dimensionless coupling receives its canonical contribution. For a coupling g_i whose mass dimension is $[g_i]$, the canonical term is $-[g_i]\, g_i$; the non-vanishing entries are

$$\beta_{g_N} \supset +2\, g_N, \qquad \beta_{\lambda_N} \supset -2\, \lambda_N, \qquad \beta_{m_i^2} \supset -2\, m_i^2, \qquad \beta_{c_\bullet} \supset +2\, c_\bullet, \tag{3}$$

with $c_\bullet$ any dimension-six coordinate (the projected sextic couplings carry mass dimension -2, since the background field carries mass dimension 1). All other coordinates are dimensionless ($[g_i] = 0$).

The no-fabrication control is implemented at the code level: assembling the full 41-vector beta function without a complete projected quantum contribution raises `IncompleteFlowError`; entries for which no traced projection exists are `NaN`, never silently zero. A valid fixed-point claim must make all 41 residuals small, and the stability matrix must be the Jacobian of the declared flow with a stated critical-exponent sign convention.

2.4 Residuals, stability matrices, and critical exponents

All stability results in this paper use the numerical central-difference Jacobian of the declared flow, evaluated at the point of interest with step δ; the critical exponents are the eigenvalues of the Jacobian with the convention

$$\theta = -\mathrm{eig}(\partial\beta_i/\partial g_j)\,, \tag{4}$$

so that a positive real part means a relevant (UV-attractive) direction. Fixed-point residuals are quoted as $|\beta_i|$ at the point.

3 Verified building blocks

3.1 Einstein–Hilbert gravity coupled to Standard-Model matter (DEP block)

The gravity–matter block follows Donà, Eichhorn and Percacci [6] (arXiv:1311.2898). Its convention label is `DEP-TYPEII`: type-II cutoff with shape parameters $\alpha = \rho = 1$, the table ghost factor $2\,2^{[d/2]}$, and minimal coupling of matter. The block uses the scalar–fermion–gauge content of the Standard Model ($n_s = 4$ Higgs real components, n_f fermions, n_v gauge fields); in this reconstruction $n_s = 4$, $n_f = 45/2$ per the standard count, and $n_v = 12$, plus the two real singlets. The anomalous dimensions of the DEP block are the coupled solution $\eta_h, \eta_\phi, \eta_E, \eta_V$ of the source equations.

In the scaffold coordinates of this project, $g_N = \tilde{G}$ and $\lambda_N = 2\tilde{\Lambda}$, so

$$\beta_{g_N} = \beta_{\tilde{G}}, \qquad \beta_{\lambda_N} = 2\,\beta_{\tilde{\Lambda}}, \tag{5}$$

where $(\tilde{\Lambda}, \tilde{G}) = (\Lambda k^{-2}, G k^{d-2})$ are the couplings of Ref. [6]. The block reproduces the published pure-gravity fixed point $(\tilde{\Lambda}, \tilde{G}) = (0.01046, 0.7725)$ and the Standard-Model row $(\tilde{\Lambda}, \tilde{G}) = (-2.399, 1.762)$. In the unimproved (table) treatment used here the Standard-Model fixed point is

$$\tilde{\Lambda}_* = -2.3987548, \qquad \tilde{G}_* = 1.7621271, \qquad \theta = (3.9609,\ 1.6436), \tag{6}$$

with $\eta_h \simeq 0.269$ and $\eta_s \simeq -0.076$ at the fixed point. The improved (self-consistent η) treatment yields $(\tilde{\Lambda}_*, \tilde{G}_*) = (-2.4644, 1.7951)$ and $\theta = (3.9988, 1.6377)$; the table convention is used throughout this paper because it is the convention under which the source's published exponent table is reproduced. All numbers are regulator- and gauge-dependent.

3.2 The Litim Einstein–Hilbert benchmark

As a cross-check of the fixed-point machinery, the pure Einstein–Hilbert truncation with the optimized cutoff [4, 5] is implemented in its own normalization ($g_{eq} = c_d\, g_N$, $c_4 = 24\pi$, $\alpha \to \infty$). The known result is reproduced exactly:

$$\lambda_N^* = \frac{1}{4}, \qquad g_N^* = \frac{1}{1536\pi}, \qquad \theta = \frac{5}{3} \pm \frac{i\sqrt{167}}{3}. \tag{7}$$

This block is kept deliberately separate from the DEP convention; its only role is to certify the Jacobian and eigenvalue pipeline.

3.3 Nonminimal scalar–curvature couplings (Narain–Percacci block)

The nonminimal couplings ξ_H, ξ_Φ, ξ_E follow the closed-form five-coupling scalar–tensor system of Narain and Percacci [7] (arXiv:0911.0386), convention label `NP-DE-DONDER-TYPEA`: $d = 4$, de-Donder gauge, optimized cutoff, one real scalar. The couplings are defined by

$$V = k^d \sum_n \lambda_{2n}\, \tilde{\phi}^{2n}, \qquad F = k^{d-2} \sum_n \xi_{2n}\, \tilde{\phi}^{2n}, \qquad \tilde{\phi} = \phi\, k^{-(d-2)/2}, \tag{8}$$

so that in $d = 4$, $\lambda_2 = m^2/k^2$ and λ_4 is the dimensionless quartic coefficient of ϕ^4. The system is implicit: the anomalous dimension $\eta = \partial_t \xi_0/\xi_0$ is fixed by the ξ_0 flow, and $\partial_t \xi_2$ appears linearly in the λ_2, ξ_2 and λ_4 flows. Writing $\beta_{\xi_0} = X + \eta\, Y$, the implicit condition $\partial_t \xi_0 = (\eta - 2)\xi_0$ is solved algebraically as

$$\eta = \frac{X + 2\xi_0}{\xi_0 - Y}, \tag{9}$$

with (λ_0, ξ_0 the dimensionless R and $R\phi^2$ couplings of the paper)

$$X = -2\xi_0 + \frac{1}{384\pi^2}\left[25 - \frac{4}{1+2\lambda_2} - \frac{24\,\xi_2}{(1+2\lambda_2)^2} + \frac{8\xi_0\,(7\xi_0 - 2\lambda_0)}{(\xi_0 - \lambda_0)^2}\right], \tag{10}$$

$$Y = \frac{1}{1152\pi^2}\,\frac{17\xi_0^2 + 18\xi_0\lambda_0 - 15\lambda_0^2}{(\xi_0 - \lambda_0)^2}, \tag{11}$$

and

$$\beta_{\lambda_0} = -4\lambda_0 + \frac{1}{32\pi^2}\left[2 + \frac{1}{1+2\lambda_2} + \frac{6\lambda_0}{\xi_0 - \lambda_0}\right] + \frac{\eta}{96\pi^2}\,\frac{5\xi_0 - 2\lambda_0}{\xi_0 - \lambda_0}, \tag{12}$$

$$\beta_{\xi_0} = -2\xi_0 + X + \eta\, Y. \tag{13}$$

The remaining three flows β_{λ_2}, β_{ξ_2}, β_{λ_4} are closed forms in $(\lambda_0, \xi_0, \lambda_2, \xi_2, \lambda_4, \eta)$; the ξ_2 flow is solved for the implicit $\partial_t \xi_2$ term,

$$\beta_{\xi_2} = \frac{Z + \eta\, W}{1 - V/\xi_0}, \tag{14}$$

and β_{λ_2}, β_{λ_4} contain the solved $\partial_t \xi_2$ combination. The full forms are listed in App. B.

The two-coupling gravity sector (GMFP) of the paper is reproduced exactly:

$$\lambda_0^* = 0.0086200674, \qquad \xi_0^* = 0.0237546805, \qquad \theta = 2.1426 \pm 2.8795\, i, \tag{15}$$

with eigenvector $(0.3557 \pm 0.3776\, i,\ 0.8549)$, obtained from the numerical Jacobian of (9)–(13); $\eta = 2$ at the fixed point. The paper's own closed-form stability-matrix entries M_{00} do not equal the

numerical derivatives of the appendix flows at the GMFP ($M_{\lambda_0\lambda_0}$ differs by exactly $+4$, $M_{\xi_0\xi_0}$ by ≈ -2.14); the published exponents and eigenvectors are reproduced by the flow-derived Jacobian, and the paper itself notes transcription errors in its precursor. We therefore use the closed-form flows and numerical Jacobians only, and record the discrepancy.

The scaffold wiring evaluates the ξ_2 flow per species on the GMFP background of (15) with its exact fixed-point value $\eta = 2$, and maps

$$\lambda_2 = m_i^2, \qquad \xi_2 = \xi_i, \qquad \lambda_4 = \lambda_i/4 \qquad (i = H, \Phi, E), \tag{16}$$

from $V = \lambda_i/4\,(\phi^2)^2$ for the radial field. Because the background is fixed at the NP GMFP, the ξ entries do not mix the NP gauge with the DEP gravity block; the two convention labels never coexist in one solve. At Gaussian matter ($\lambda_2 = \xi_2 = \lambda_4 = 0$) the ξ flow vanishes (residual $\sim 10^{-16}$), so $\xi_i^* = 0$ is an exact fixed point of this block.

3.4 The scalar local-potential block: masses, quartics, wave functions

The six real scalars (four Higgs components, Φ_c, E) are described by a local potential $U_k(h, \Phi, E)$ whose Hessian on a constant background φ is

$$m_{ab}^2(\varphi) = m_{ab}^2(0) + \tfrac{1}{2}\,\lambda_{abcd}\,\varphi_c\varphi_d + \tfrac{1}{24}\,c_{abcdef}\,\varphi_c\varphi_d\varphi_e\varphi_f, \tag{17}$$

where λ_{abcd} is the quartic tensor and c_{abcdef} the sextic tensor of App. C. With the optimized regulator (2), the scalar contribution to the flow of the potential is the constant-background trace

$$\partial_t U_k\big|_{\text{scalar}} = \frac{k^4}{32\pi^2}\sum_{a=1}^{6}\Big(1 + \frac{m_a^2(\varphi)}{k^2}\Big)^{-1}, \tag{18}$$

with m_a^2 the eigenvalues of (17). Projecting at zero field gives the quantum contributions to the masses and quartics,

$$\beta_{m_i^2} \supset \frac{\partial^2\partial_t U_k}{\partial\varphi_i^2}\Big|_0, \qquad \beta_{\lambda_i} \supset \frac{1}{6}\,\frac{\partial^4\partial_t U_k}{\partial\varphi_i^4}\Big|_0, \qquad \beta_{\kappa_{ij}} \supset \frac{\partial^4\partial_t U_k}{\partial\varphi_i^2\,\partial\varphi_j^2}\Big|_0, \tag{19}$$

computed with mixed finite-difference stencils whose step is checked against an analytic toy derivative. The Gaussian point is an exact fixed point of this block, and the fourth-order projections match the analytically known one-scalar flow. The wave-function coordinates are driven by the DEP anomalous dimensions of Sec. 3.1,

$$\beta_{Z_i} = -\eta_s\, Z_i \qquad (i = h, \phi, E), \tag{20}$$

with $\eta_s \simeq -0.076$ at the Standard-Model fixed point, so $Z_i^* = 0$.

3.5 Dimension-six scalar potentials (sextic LPA)

The ten scalar dimension-six potentials are projected from the same trace (18) with the sixth-order potential. The normalization convention assigns a factor $1/2^{a+b+c}$ to every monomial $(h^2)^a(\Phi^2)^b(E^2)^c$ with $a+b+c = 3$, the same "two per invariant factor" rule that fixes the quartic sector ($\lambda_H/4\,(h^2)^2$, $\kappa_{H\Phi}/4\,(h^2)\Phi^2$). The projections are

$$c_{H6} = \frac{1}{90}\,\frac{\partial^6\partial_t U_k}{\partial h^6}\Big|_0, \qquad c_{H4\Phi2} = \frac{1}{6}\,\frac{\partial^4\partial^2\partial_t U_k}{\partial h^4\,\partial\Phi^2}\Big|_0, \qquad c_{H2\Phi2E2} = \frac{\partial^2\partial^2\partial^2\partial_t U_k}{\partial h^2\,\partial\Phi^2\,\partial e^2}\Big|_0, \tag{21}$$

and cyclically; the $O(4)$ combinatorics of the Higgs sector are fixed by the sextic tensor of App. C, which is validated by a unit test that reconstructs every monomial of the potential from the tensor at generic couplings. The finite-difference step is chosen in the convergence plateau ($\delta \simeq 5 \times 10^{-2}$; values at $\delta = 3 \times 10^{-2}$ and 7×10^{-2} agree to a few percent). The block passes three controls: it reduces exactly to the quartic flow at vanishing sextics; the pure-toy derivatives $\partial^6/\partial h^6 = 720$, $\partial^4\partial^2 = 144$, $\partial^2\partial^2\partial^2 = 48$ are reproduced; and all ten entries vanish at the Gaussian point to the stencil noise floor ($\sim 10^{-11}$).

3.6 The curvature-squared sector and its negative result

The R^2 sector follows Codello, Percacci and Rahmede [8] (arXiv:0805.2909), one-loop approximation with the type-III generalized optimized cutoff and the Shapiro gauge choice that makes the quadratic action minimal fourth order. The action ansatz is

$$S = \int \sqrt{g}\,\Big[\frac{1}{2\lambda}\,C^2 + \frac{1}{\xi}\,R^2 - \frac{1}{\rho}\,E\Big], \tag{22}$$

with C^2 the Weyl-squared, E the Euler density, and beta functions

$$\beta_\lambda = -\frac{133}{10}\frac{\lambda^2}{(4\pi)^2}, \qquad \beta_\xi = -\frac{10\lambda^2 - 5\lambda\xi + 5\xi^2/36}{(4\pi)^2}, \qquad \beta_\rho = \frac{196}{45}\frac{\rho^2\lambda}{(4\pi)^2}. \tag{23}$$

In the ratios $\omega = -3\lambda/\xi$ and $\theta = \lambda/\rho$ the fixed points are $\omega_* \in \{-5.467, -0.0228\}$ and $\theta_* = 56/171 \simeq 0.3275$; the ω conversion is exactly consistent with (23) via the chain rule (checked algebraically and numerically). We record the paper-internal caveat that the printed β_θ does not follow from β_ρ by the chain rule; the θ sector is decoupled at this order, so the discrepancy affects no wired coordinate, and $\theta_* = 56/171$ is used as printed.

The scaffold coordinates are the inverse couplings

$$a_{R^2} = \frac{1}{\xi}, \qquad a_{C^2} = \frac{1}{2\lambda}, \tag{24}$$

defined only for $\lambda \neq 0$ (singular at the Gaussian point). From (23),

$$\beta_{a_{C^2}} = \frac{133}{320\,\pi^2} \approx 0.04211\ (>0,\ \text{constant}), \qquad \beta_{a_{R^2}} = \frac{10x^2 - 5x + 5/36}{(4\pi)^2}, \qquad x = \frac{a_{R^2}}{2a_{C^2}}. \tag{25}$$

$\beta_{a_{R^2}}$ vanishes on the two lines $a_{R^2} = 2x\,a_{C^2}$ with $x \in \{0.0295207,\ 0.4704793\}$, but $\beta_{a_{C^2}}$ is strictly positive and constant, so the Weyl-squared direction runs monotonically and the (a_{R^2}, a_{C^2}) subsystem has no fixed point in this parametrization. We record this as a negative result, not a papered-over gap. Physically, the underlying flow (23) has its Gaussian point at $\lambda = \xi = 0$ (the Weyl-squared coupling is asymptotically free in the one-loop approximation), and the inverse parametrization diverges there; the conclusion is a statement about the scaffold coordinates, which are the ones used in the 41-coordinate basis.

4 The joint fixed point

4.1 Fixed-point vector and residuals

Assembling the blocks of Sec. 3 into `beta_partial` (33 of the 41 entries finite; the remaining 8 are *pending*, Sec. 5), the joint fixed point of the derived flow is

$$(\lambda_N,\, g_N) = (-4.7975095,\ 1.7621271), \qquad \text{all other 31 derived coordinates } = 0, \tag{26}$$

i.e. the DEP Standard-Model row (6) with the matter sector at its Gaussian point. The residuals at (26) are: gravity and matter entries zero to machine precision; ξ_i entries $\sim 10^{-16}$ (exact fixed point of the NP block, Sec. 3.3); dimension-six entries $\sim 10^{-11}$ (the finite-difference noise floor of the sextic projections). The joint fixed point is therefore a genuine fixed point of the derived flow to the numerical accuracy of the projections.

4.2 The 33-dimensional stability spectrum

The stability matrix is the numerical central-difference Jacobian of the derived flow restricted to the 33 derived coordinates, evaluated at (26). Two conventions are reported: the quantum-only Jacobian of `beta_partial`, and the quantum-plus-canonical Jacobian with the canonical terms (3) added for the mass-squared and dimension-six entries (the DEP gravity block contains its own canonical terms; all other matter couplings are dimensionless). The quantum-plus-canonical spectrum is the physically relevant one and is displayed in Table 1.

Table 1: Quantum-plus-canonical critical exponents $\theta = -\mathrm{eig}(\partial\beta/\partial g)$ of the 33-dimensional derived flow at the joint fixed point (26).

θ	multiplicity	class	directions
+3.9609	1	relevant	Λ-direction (λ_N)
+2.8554	3	relevant	m_H^2, m_Φ^2, m_E^2
+2.0000	3	relevant	ξ_H, ξ_Φ, ξ_E
+1.6436	1	relevant	$\tilde{G}$-direction (g_N)
0.0000	12	marginal	$g_1, g_2, g_3, y_t, y_b, y_\tau$ and the six quartics
−0.0756	3	irrelevant	Z_h, Z_ϕ, Z_E
−2.0000	10	irrelevant	the ten dimension-six potentials

The two gravity eigenvalues reproduce the DEP table-convention exponents $(3.961, 1.644)$ of (6) exactly. The counting is

$$\underbrace{8}_{\text{relevant}} + \underbrace{12}_{\text{marginal}} + \underbrace{13}_{\text{irrelevant}} = 33. \tag{27}$$

4.3 Interpretation

The spectrum has a transparent block structure.

- *Gravity.* The two exponents +3.9609 and +1.6436 are the DEP eigenvalues, unchanged by the (Gaussian) matter sector; they confirm that the fixed point is UV attractive in the Einstein–Hilbert directions.
- *Mass-squared directions.* $\theta = 2.8554 = 2 + 0.8554$: the canonical weight +2 (from $-[m_i^2] = +2$ per (3), i.e. $\beta_{m^2} \supset -2m^2$) is augmented by a quantum shift from the scalar one-loop traces of Secs. 3.4–3.5 evaluated at the Standard-Model row of the DEP block. The three masses are relevant with identical exponents.
- *Nonminimal couplings.* $\theta = 2.0000$ exactly for all three ξ_i: at the NP GMFP background with $\eta = 2$ the ξ_2 flow (14) has an exact linear coefficient, and the ξ_i directions inherit it. This is a pure quantum result; the ξ_i have canonical weight zero.
- *Marginal matter couplings.* The twelve coordinates $g_1, g_2, g_3, y_t, y_b, y_\tau$ and the six quartic couplings $\lambda_H, \lambda_\Phi, \lambda_E, \kappa_{H\Phi}, \kappa_{HE}, \kappa_{\Phi E}$ have vanishing eigenvalues: at Gaussian matter every one-loop contribution is of higher order in the couplings (the quartic and Yukawa beta functions have

no linear terms in the respective couplings, and the gauge beta functions vanish at zero gauge coupling), and the canonical weights are zero. The stability matrix does not lift their degeneracy at this point, a well-known feature of Gaussian matter sectors.

- *Wave functions.* $\theta = -0.0756 = -\eta_s$ with $\eta_s \simeq -0.076$ from (20); the Z_i are irrelevant and the fixed point is $Z_i^* = 0$.
- *Dimension-six potentials.* $\theta = -2.0000$ exactly: pure canonical scaling (3); the quantum parts vanish at Gaussian matter to the stencil floor, so all ten operators are irrelevant at this fixed point.

The structure is consistent with the expectation for an interacting gravity–matter fixed point with Gaussian matter: the gravity sector controls the flow, the masses and nonminimal couplings are relevant, and all higher-dimensional scalar operators are irrelevant.

5 Coverage, limitations, and pending directions

Thirty-three of the 41 coordinates are derived and verified:

- `beta_partial` (quantum-only, DEP gravity + SM matter + scalar LPA + NP ξ + sextic LPA): entries 0–1 (gravity), 5–19 (SM gauge/Yukawa/quartics/portals), 20–22 (ξ_i), 23–25 (Z_i), 26–35 (dimension-six potentials), plus the scalar LPA mass entries 11,13,15.
- The remaining coordinates are *pending* and return `NaN`: a_{R^2}, a_{C^2} (handled by the negative result of Sec. 3.6); the derivative operators $c_{H\Box}, c_{\Phi\Box}, c_{E\Box}$ (not projectable by a constant-background LPA); and the gauge-strength portals $c_{HF^2}, c_{\Phi F^2}, c_{EF^2}$ (require a dressed-gauge-sector trace).

No flow value is fabricated for the pending coordinates, and the framework refuses to present a 41-dimensional fixed-point claim. Every numerical result is accompanied by its convention label, and the unit-test suite pins each verified number.

6 Discussion

The reconstruction illustrates a methodological point that is often understated: in the FRG, a coupling is not "there" until its projected flow is written down. The 41-coordinate claim that motivated this project contains no operator list, no beta functions, no fixed-point vector, and no stability matrix; we have therefore delivered a smaller system (33 derived coordinates) in which every object is written down, sourced, and verified. The main physics findings are:

1. The joint fixed point of the derived flow is the DEP Standard-Model fixed point with Gaussian matter. Its spectrum has exactly 8 relevant, 12 marginal, and 13 irrelevant directions. The gravity eigenvalues are the DEP values $(3.961, 1.644)$; the masses and nonminimal couplings are relevant; the ten dimension-six scalar operators are irrelevant by canonical scaling alone.
2. The ξ_i directions are exact: the Narain–Percacci GMFP background with $\eta = 2$ forces $\theta = 2$ and an exact Gaussian fixed point of the nonminimal sector. This is a clean, verifiable sub-result and a nontrivial consistency check between the two published blocks.
3. The curvature-squared sector has no fixed point in the scaffold parametrization (24): $\beta_{a_{C^2}} > 0$ constant. In the original couplings the Weyl-squared direction is asymptotically free toward the Gaussian point; the negative result is a property of the inverse coordinates used in the basis. We record it explicitly, because an unguarded reader of the scaffold might otherwise search for a fixed point that does not exist.
4. The twelve marginal matter couplings (gauge, Yukawa, quartic) at Gaussian matter are not a prediction of fixed values; they are the statement that the stability matrix is degenerate in those directions at the Gaussian point. A fully interacting matter fixed point would require the

coupled matter beta functions away from the Gaussian point, which is outside the currently derived system.

Limitations. (i) All gravity and matter blocks are one-loop (threshold-functional) or LPA-level; there is no fermion trace in the scalar projections, no dressed gauge sector, and no derivative (curvature-squared or box-type) projection for matter. (ii) The pending coordinates are not fabricated; the joint fixed point is a statement about the 33 derived coordinates only. (iii) The conventions (type-II vs. optimized regulator, de-Donder vs. type-A gauge, table vs. improved ghost factors) are deliberately separated; mixing them would produce numbers that are neither paper's. (iv) The NP M_{00} discrepancy and the CPR β_θ inconsistency are recorded paper-internal observations; both are resolved in favor of the closed-form flows and decoupled sectors.

7 Conclusions

Within a fully declared truncation, the reconstructed gravity–matter system has a UV-attractive joint fixed point with 8 relevant, 12 marginal, and 13 irrelevant directions at the DEP Standard-Model row with Gaussian matter. All 33 derived flows are traced to published sources or documented projections and are pinned by unit tests; the 8 remaining coordinates are explicitly pending, and the curvature-squared sector is documented as having no fixed point in the scaffold parametrization. The accompanying implementation (`outputs/uv_reconstruction_41/`) is the precise definition of the calculation; this paper is the human-readable record of it.

Reproducibility

The complete implementation lives in `outputs/uv_reconstruction_41/`. The verified numbers of this paper are regenerated by

```
$PY -m unittest discover -s tests
```

(44 tests), with `PY` any Python 3.10+ interpreter with NumPy. The joint fixed point and spectrum are produced by `joint_fixed_point_stability.py`; the convention register is `DECLARED_TRUNCATION_V0_1.md`.

References

[1] C. Wetterich, *Exact evolution equation for the effective potential*, Phys. Lett. B 301 (1993) 90, arXiv:hep-ph/9211263.

[2] J. Berges, N. Tetu, C. Wetterich, *The effective average action and its flow equation*, Phys. Rept. 363 (2002) 223, arXiv:hep-ph/0005122.

[3] M. Reuter, *Nonperturbative evolution equation for quantum gravity*, Phys. Rev. D 57 (1998) 971, arXiv:hep-th/9609127.

[4] D. F. Litim, *Optimised renormalisation group flows*, Phys. Rev. D 64 (2001) 105007, arXiv:hep-th/0009045.

[5] D. F. Litim, *Fixed points of quantum gravity and the renormalisation group*, Phys. Rev. D 76 (2007) 125001, arXiv:hep-th/0606044.

[6] P. Donà, A. Eichhorn, R. Percacci, *Matter matters in asymptotically safe quantum gravity*, Phys. Rev. D 89 (2014) 084035, arXiv:1311.2898.

[7] G. Narain, R. Percacci, *On the stability of the asymptotic solutions of the Asymptotic Safety program*, Class. Quantum Grav. 27 (2010) 075001, arXiv:0911.0386.

[8] A. Codello, R. Percacci, C. Rahmede, *Investigating the ultraviolet properties of gravity with a Wilsonian renormalization group equation*, Ann. Phys. (NY) 324 (2009) 414, arXiv:0805.2909.

[9] R. Percacci, *An Introduction to Covariant Quantum Gravity and Asymptotic Safety*, World Scientific (2017).

A The declared 41-coordinate operator basis

Table 2: The declared 41-coordinate basis. $[g_i]$ is the mass dimension of the coupling; the canonical term is $-[g_i]\, g_i$.

#	coordinate	$[g_i]$	operator
1	g_N	-2	R (dimensionless Newton coupling)
2	λ_N	$+2$	cosmological term
3	a_{R^2}	$-$	R^2
4	a_{C^2}	$-$	Weyl-squared
5–7	g_1, g_2, g_3	0	SM gauge kinetic terms
8–10	y_t, y_b, y_τ	0	third-generation Yukawas
11,13,15	m_H^2, m_Φ^2, m_E^2	$+2$	scalar masses
12,14,16	$\lambda_H, \lambda_\Phi, \lambda_E$	0	scalar quartics
17	$\kappa_{\Phi E}$	0	$\Phi_c^2 E^2$
18–19	$\kappa_{H\Phi}, \kappa_{HE}$	0	$\lvert H\rvert^2\Phi_c^2$, $\lvert H\rvert^2 E^2$
20–22	ξ_H, ξ_Φ, ξ_E	0	$R\lvert H\rvert^2$, $R\Phi_c^2$, RE^2
23–25	Z_h, Z_ϕ, Z_E	0	wave-function renormalizations
26	c_{H6}	-2	$\lvert H\rvert^6$
27	$c_{\Phi 6}$	-2	Φ^6
28	c_{E6}	-2	E^6
29–34	$c_{H4\Phi 2}, \ldots, c_{\Phi 2E4}$	-2	mixed sixth-order potentials
35	$c_{H2\Phi 2E2}$	-2	$\lvert H\rvert^2\Phi^2 E^2$
36–38	$c_{H\Box}, c_{\Phi\Box}, c_{E\Box}$	-2	$\lvert H\rvert^2\Box\lvert H\rvert^2$, $\Phi^2\Box\Phi^2$, $E^2\Box E^2$ (pending)
39–41	$c_{HF^2}, c_{\Phi F^2}, c_{EF^2}$	-2	$\lvert H\rvert^2F^2$, Φ^2F^2, E^2F^2 (pending)

B The Narain–Percacci five-coupling system

With $\lambda_0, \xi_0, \lambda_2, \xi_2, \lambda_4$ as in Sec. 3.3, η from (9), $D \equiv \xi_0 - \lambda_0$, $A \equiv 1 + 2\lambda_2$, $B \equiv 1 + 2\xi_2$, and $\dot{\xi}_2 \equiv \beta_{\xi_2}$, the transcribed closed forms are

$$\begin{aligned} Z = \frac{1}{576\pi^2}\Bigg[& \frac{A}{D}\Big(9 + \frac{39\xi_0}{D} + \frac{60\xi_0^2}{D^2}\Big) + \frac{3(3+32\xi_2)}{D} - \frac{6\xi_0(11+2\xi_2)}{D^2} - \frac{60\xi_0^2 B}{D^3} + \frac{216\xi_2 B^2}{A^3 D} \\ & + \frac{9\big(\lambda_0(5-2\xi_2) - 2\xi_0 B\big)B}{AD^2} + \frac{27B(1-10\xi_2-16\xi_2^2)}{A^2 D} + \frac{108\xi_0\xi_2 B^2}{A^2 D^2} \\ & + \frac{72\lambda_4(1+12\xi_2+2\lambda_2)}{A^3}\Bigg], \end{aligned} \tag{28}$$

$$\begin{aligned} W = \frac{1}{1152\pi^2}\Bigg[& \frac{A}{D}\Big(3 + \frac{18\xi_0}{D} + \frac{20\xi_0^2}{D^2}\Big) + \frac{15\xi_2}{\xi_0} - \frac{6(1+\xi_2)}{D} - \frac{10\xi_0(3+4\xi_2)}{D^2} - \frac{20\xi_0^2 B}{D^3} \\ & - \frac{3\big(\lambda_0 - \xi_0(5-4\xi_2)\big)B}{AD^2} + \frac{36\xi_0\xi_2 B^2}{A^2 D}\Bigg], \end{aligned} \tag{29}$$

$$\begin{aligned} V = {} & \frac{1}{1152\pi^2}\Bigg[-15 + \frac{54\xi_0}{D} + \frac{20\xi_0^2}{D^2} - \frac{6\xi_0(7+2\xi_2)}{AD}\Bigg] \\ & - \frac{1}{1152\pi^2}\cdot\frac{144\,\xi_0\xi_2\,B}{A^2 D}, \end{aligned} \tag{30}$$

with β_{ξ_2} the solved implicit form

$$\beta_{\xi_2} = \frac{Z + \eta W}{1 - V/\xi_0}, \tag{31}$$

and the remaining flows

$$\begin{aligned} \beta_{\lambda_2} = {} & -2\lambda_2 + \frac{1}{48\pi^2}\Bigg[\frac{9\lambda_0 B}{2D^2} - \frac{9(2\lambda_0 - \xi_0)B^2}{2AD^2}\Bigg] \\ & - \frac{1}{48\pi^2}\Bigg[\frac{9B^2}{2A^2 D} + \frac{18\lambda_4}{A^2}\Bigg] \\ & + \frac{\eta}{96\pi^2}\Bigg[-\frac{2\xi_2}{\xi_0} + \frac{3\xi_0 B}{2D^2} - \frac{3\xi_0 B^2}{2AD^2}\Bigg] + \frac{\dot{\xi}_2}{\xi_0}\cdot\frac{1}{96\pi^2}\Bigg[2 - \frac{3\xi_0}{D} + \frac{6\xi_0 B}{AD}\Bigg], \end{aligned} \tag{32}$$

$$\begin{aligned} \beta_{\lambda_4} = {} & \frac{1}{48\pi^2}\Bigg[\frac{9}{4D^2}\Big(5A(1+4\xi_2) - B(21+62\xi_2) + \frac{33B^3}{A} - \frac{B^3(23+24\xi_2)}{A^2} + \frac{6B^4}{A^3}\Big) \\ & + \frac{9\xi_0(\xi_2-\lambda_2)^2}{D^3}\,q - \frac{72\lambda_2\lambda_4 B(1-4\lambda_2+6\xi_2)}{DA^3} + \frac{9\xi_0\lambda_4\,q}{D^2} + \frac{216\lambda_4^2}{A^3}\Bigg] \\ & + \frac{\eta}{96\pi^2}\Bigg[\frac{2\xi_2^2}{\xi_0^2} + \frac{3\xi_0(\xi_2-\lambda_2)^2}{D^3}\,q + \frac{3\xi_0\lambda_4\,q}{D^2}\Bigg] \\ & + \frac{\dot{\xi}_2}{\xi_0}\cdot\frac{1}{96\pi^2}\Bigg[-\frac{2\xi_2}{\xi_0} - \frac{3\xi_0(\xi_2-\lambda_2)}{D^2}\Big(12\frac{B^2}{A^2} - 21\frac{B}{A} + 10\Big) - \frac{24\xi_0\lambda_4(1-4\lambda_2+6\xi_2)}{A^2 D}\Bigg], \end{aligned} \tag{33}$$

where

$$q = 6\frac{B^2}{A^2} - 10\frac{B}{A} + 5. \tag{34}$$

These equations reproduce the paper's GMFP, exponents, and eigenvector (Sec. 3.3).

C The sextic tensor and its projections

With the potential written $V = c_{abcdef}\,\phi^a\phi^b\phi^c\phi^d\phi^e\phi^f/6!$ and the monomial convention of Sec. 3.5, the $O(4)\times O(1)\times O(1)$ sextic tensor is generated from the ten coordinates by the following symmetric assignments (indices $i,j,k \in \{1..4\}$ label Higgs components, $\Phi = 5$, $E = 6$; each line is assigned to every distinct index pattern):

$$c_{iiiiii} = 90\,c_{H6}, \qquad c_{\Phi\Phi\Phi\Phi\Phi\Phi} = 90\,c_{\Phi 6}, \qquad c_{EEEEEE} = 90\,c_{E6}, \tag{35}$$

$$c_{iiiijj} = 18\,c_{H6}, \qquad c_{iijjkk} = 6\,c_{H6}, \tag{36}$$

$$c_{iiii\Phi\Phi} = 6\,c_{H4\Phi 2}, \qquad c_{iijj\Phi\Phi} = 2\,c_{H4\Phi 2}, \qquad c_{iiiiEE} = 6\,c_{H4E2}, \qquad c_{iijjEE} = 2\,c_{H4E2}, \tag{37}$$

$$c_{ii\Phi\Phi\Phi\Phi} = 6\,c_{H2\Phi 4}, \qquad c_{iiEEEE} = 6\,c_{H2E4}, \tag{38}$$

$$c_{\Phi\Phi\Phi\Phi EE} = 6\,c_{\Phi 4E2}, \qquad c_{\Phi\Phi EEEE} = 6\,c_{\Phi 2E4}, \tag{39}$$

$$c_{ii\Phi\Phi EE} = c_{H2\Phi 2E2}. \tag{40}$$

A unit test reconstructs every monomial from the tensor at generic couplings, guaranteeing the convention is enforced. The projections of the trace (18) onto the ten operators are

$$c_{H6} = \frac{1}{90}\,\frac{\partial^6}{\partial h^6}\,\partial_t U_k\Big|_0, \qquad c_{\Phi 6} = \frac{1}{90}\,\frac{\partial^6}{\partial \Phi^6}\,\partial_t U_k\Big|_0, \qquad c_{E6} = \frac{1}{90}\,\frac{\partial^6}{\partial e^6}\,\partial_t U_k\Big|_0, \tag{41}$$

$$c_{H4\Phi 2} = \frac{1}{6}\,\frac{\partial^6}{\partial h^4\partial\Phi^2}\partial_t U_k\Big|_0, \qquad \text{etc.}, \qquad c_{H2\Phi 2E2} = \frac{\partial^6}{\partial h^2\partial\Phi^2\partial e^2}\partial_t U_k\Big|_0, \tag{42}$$

computed with mixed finite-difference stencils at step $\delta = 5\times 10^{-2}$ in the convergence plateau.